Project Report

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Introduction

Mushroom classification is an important problem in food safety because eating poisonous mushrooms can be hazardous to health or even fatal. It's sometimes hard to tell whether the mushroom is edible or poisonous since many poisonous ones have a similar appearance to safe ones. This is why it is crucial to be able to classify mushrooms and accurately identify them, particularly for individuals who eat wild mushrooms.

The goal of this project is to create a machine learning model which can be used to classify a mushroom edible or poisonous with its physical features. Features used in the dataset are cap shape, cap color, cap odor, gill size and habitat. The model can identify these characteristics and gain insights into patterns to accurately classify edible and poisonous mushrooms.

This project is motivated by the use of the data science techniques for a practical problem with great importance. Machine learning can help minimize human error and offer a quick and accurate means of classification. This project is also of use in understanding key concepts like data preprocessing, the analysis of features, and the classification algorithms making the project both useful and informative.

Data Set Description

The dataset used in this project contains 61,069 data points and 21 features, as indicated by the dataset shape (61069, 21). Each row represents a mushroom instance described by various physical characteristics.

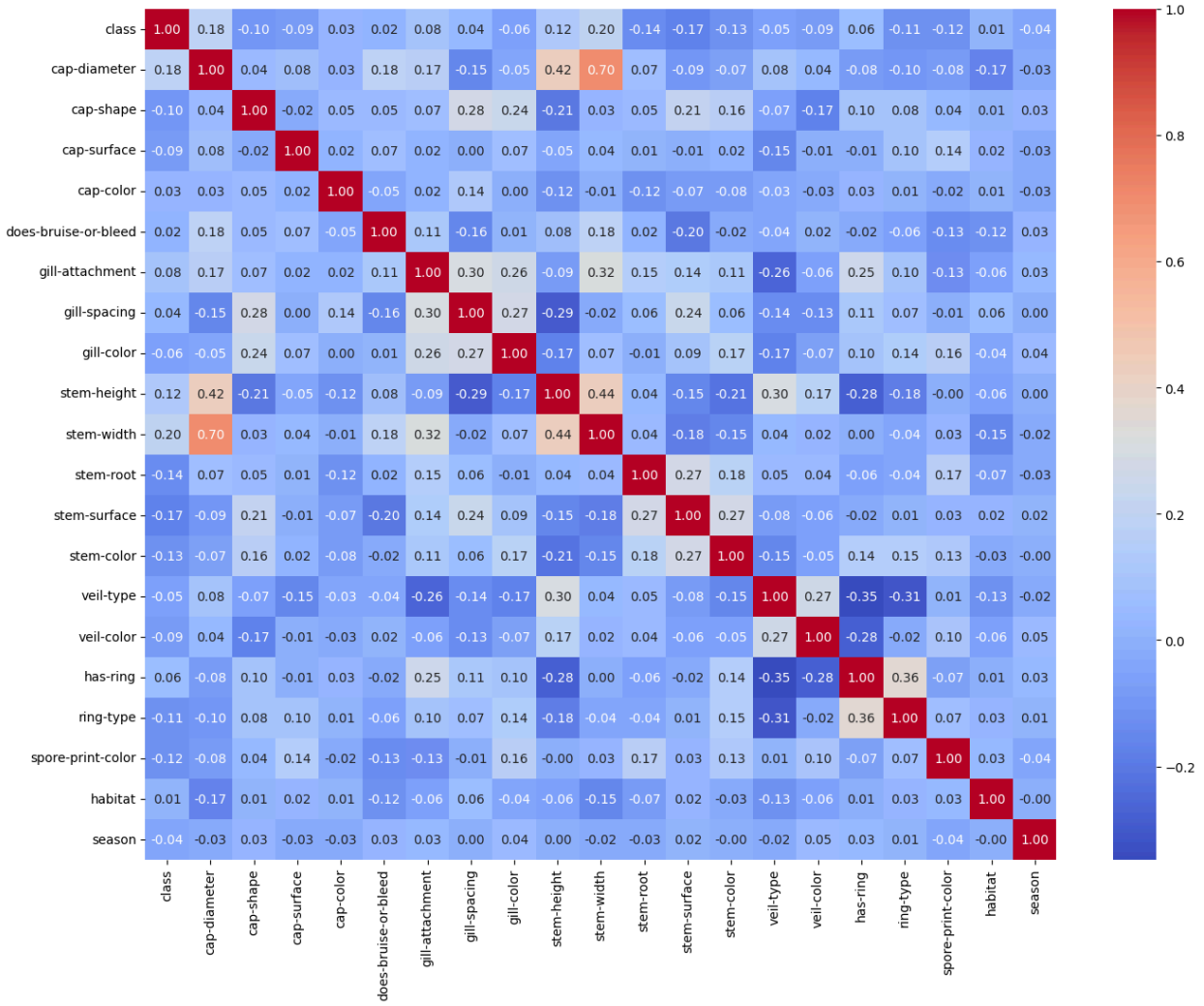
This is a classification problem because the target variable ("class") consists of discrete categories (edible or poisonous), rather than continuous numerical values. The goal is to assign each mushroom to one of these classes based on its features.

The dataset primarily consists of categorical features such as cap shape, cap color, gill color, habitat, and season. There are also a few quantitative (numerical) features, including cap diameter, stem height, and stem width.

Since most machine learning models require numerical input, categorical variables should be encoded (using one-hot encoding or label encoding). As algorithms can't work with non-numeric data directly, they need to be encoded.

There are also missing values for certain columns in the data set, including cap-surface, gill-spacing, gill-attachment, stem-surface, stem-root, veil-type, veil-color and spore-print-color, to which these must be addressed in the pre-processing phase.

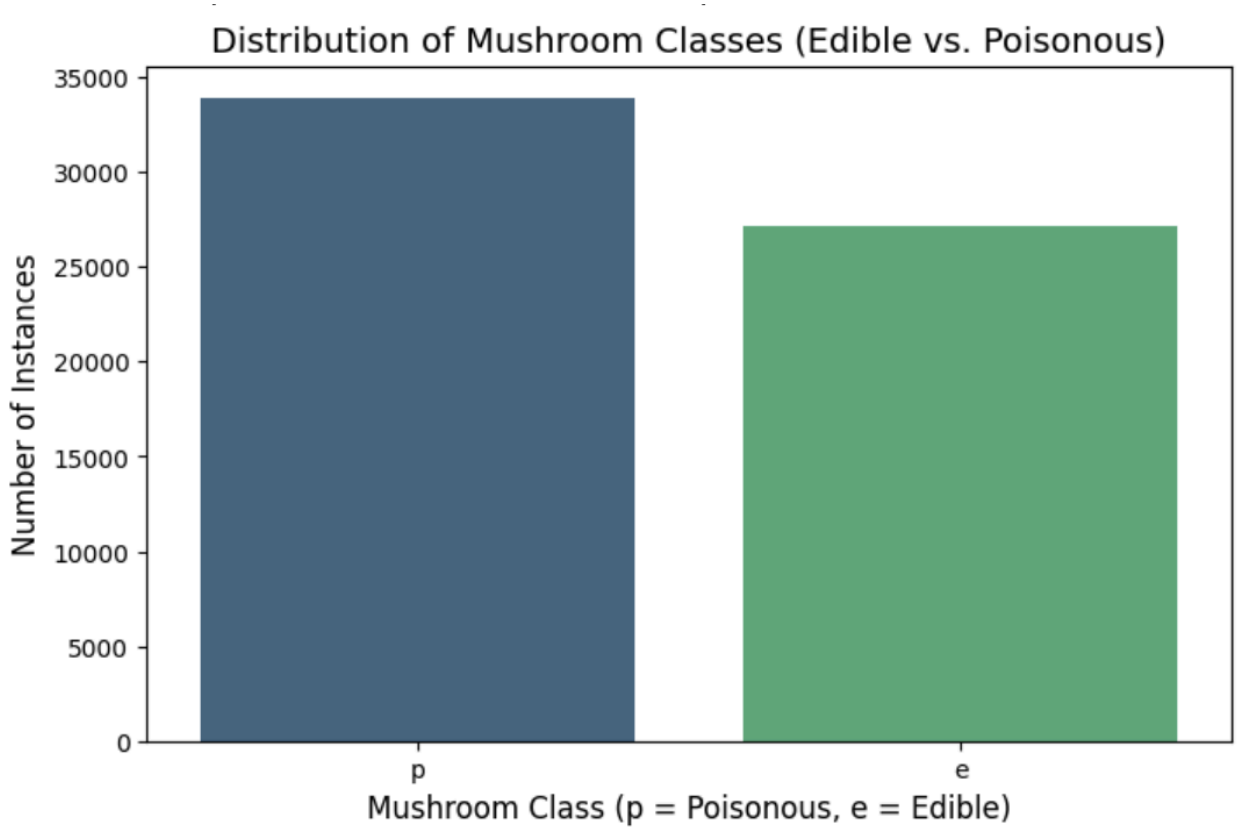
Correlation Test



The correlation analysis identifies the strongest physical predictors of mushroom toxicity by highlighting features with significant relationships to the **class** variable. It also reveals redundant features that provide overlapping information, allowing for a more streamlined and efficient model. Furthermore, the results help distinguish which specific traits, such as cap-color or gill-spacing, are statistically independent and which are highly interdependent. This process is essential for feature selection, ensuring that only the most impactful data is used to improve the accuracy of the final classification.

Distribution Test

To analyze whether the dataset is balanced, the distribution of the target variable ("class") was examined using a bar chart. The dataset contains two classes: **'p' (poisonous)** and **'e' (edible)**.



From the analysis, the number of instances in each class is:

- Poisonous (p): 33,888
- Edible (e): 27,181

This shows that the dataset is not perfectly balanced, as the number of poisonous mushrooms is higher than the number of edible ones. However, the difference is not extremely large, so the dataset can be considered **moderately imbalanced**. The bar chart visualization clearly illustrates

this imbalance, with the poisonous class having a taller bar compared to the edible class. This imbalance may slightly influence the model, making it more biased toward predicting the majority class (poisonous).

Dataset Pre-processing

In this section, we looked for the inconsistencies in our data which can affect our overall learning process of the models. After finding it, those were fixed with some data- processing techniques.

Firstly, some features such as ‘stem-root’, ‘veil-type’, ‘veil-color’, and ‘spore-print-color’ had a significant number of missing values in it (more than 75%) therefore these do not have much to contribute, instead making it complicated. Hence, these features were simply dealt using column dropping to reduce dataset complexity

Secondly, there were features such as ‘cap-diameter','season', ‘cap-color', 'gill-color', 'stem-color', ‘ring-type' ,'stem-height', 'stem-width' either redundant or not useful for prediction. It was determined through the low correlation between each feature and the target feature ‘class’ consequently resulting in lower importance.

Thirdly, our dataset also have other features such as ‘cap-shape’, ‘cap-surface’, ‘does-bruise-or-bleed’, ‘gill attachment’, ‘gill-spacing’,’ stem-surface’, ‘has-ring, ‘habitat’ which also had missing values earlier but was not removed unlike previous one. Rather we handled the

missing values through replacing using SimpleImputer with the most frequent strategy, as the data is categorical.

Now for the encoding section we had features which were categorical. Now to feed those to the model and make it understand we needed to convert those into numerical form. The Label Encoding technique was applied to binary class features such as ‘class’, ‘does-bruise-or-bleed’, and ‘has-ring’ assigning 0s and 1s.

On the other hand, One-Hot Encoding was applied to multi-category features including ‘cap-shape',‘cap-surface’,‘gill-attachment’,‘gill-spacing’, ‘stem-surface’ and 'habitat’. Here, depending on the number of distinct values in each feature containing, additional columns are added. The technique only assigns 1 to the selected column and the rests are assigned 0.
Lastly, our final selected feature data had to be scaled as well, as it creates chances of overfitting or biasness or even data leakage if the data is too large or an outlier. Or else the model will start to give more preference to it and learn. To fix it, StandardScaler was used to standardize the features.

Dataset Splitting

We splitted our dataset into training and testing sets using a 80:20 ratio.

- Training Set: 80%
- Testing Set: 20%

A fixed random state was used to ensure reproducibility.

Model Training & Testing

In this project, multiple supervised learning models were used to classify mushrooms as edible or poisonous. The dataset was split into training and testing sets to evaluate performance.

- K-Nearest Neighbors (KNN): KNN classifies data based on the majority class of its nearest neighbors. It is simple but can be slow for large datasets.

- Decision Tree: Decision Tree splits data based on feature values to make predictions. It is easy to understand but may overfit.

- Logistic Regression: Logistic Regression predicts class probabilities using a sigmoid function. It works well for binary classification with simple relationships.

- Neural Network (MLP Classifier): An MLP Classifier was used to capture complex patterns in the data. It performs well but requires parameter tuning.

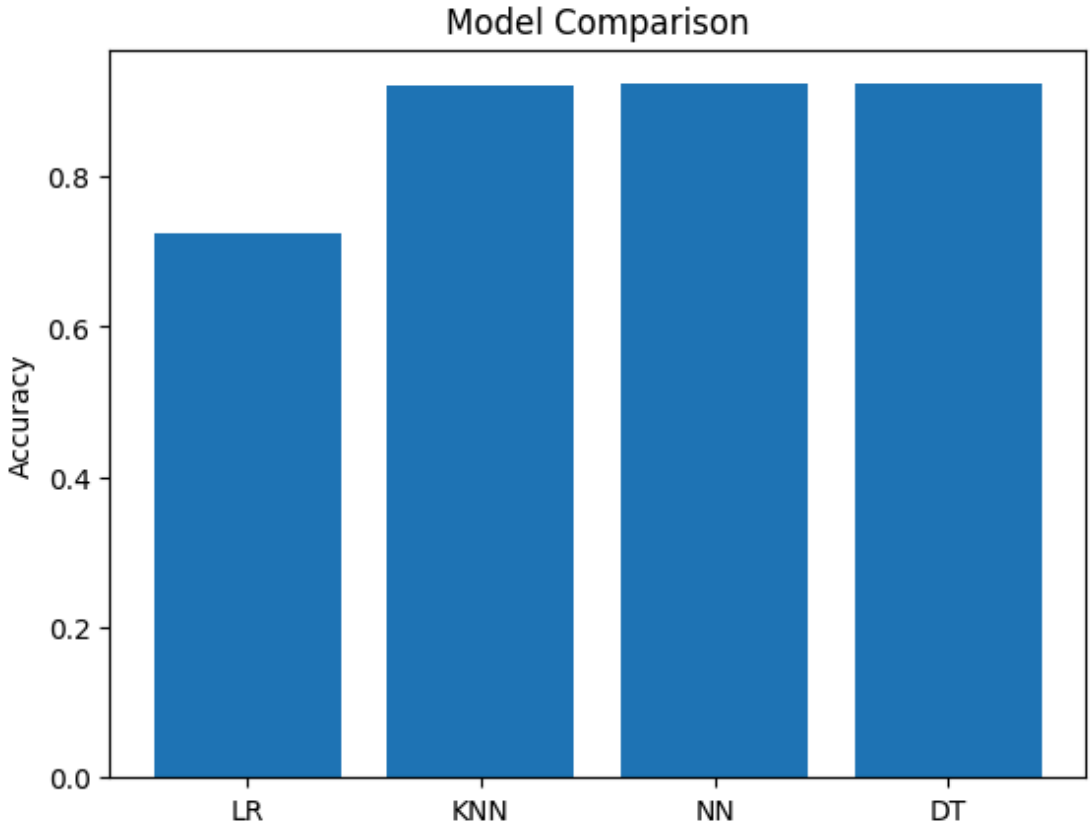
K-Means clustering

The K-Means clustering is an unsupervised algorithm which organizes the data and gathers into groups based on similarity without using actual labels. It was used with 2 clusters in this project to indicate the 2 mushroom classes(P or E). The results reveal that the data exhibits some natural clustering, but the clustering is not perfect as the algorithm does not rely on the label information.

Comparison Analysis

We compared our four machine learning models to classify mushrooms as edible or poisonous. The evaluation was based on accuracy, precision, recall, F1-score, and the Area Under the ROC Curve (AUC).

Prediction Accuracy Comparison



The bar chart illustrates the performance of each model. Decision Tree and Neural Network demonstrated the highest efficacy, followed closely by KNN.

- Decision Tree: Achieved the highest accuracy (100%).
- Neural Network: Showed exceptional performance (99.8%).
- KNN: Performed very well with an accuracy of approximately 99%.
- Logistic Regression: Recorded the lowest accuracy at approximately 88%, suggesting that a linear approach is less suited for this complex biological dataset.

Score Comparison

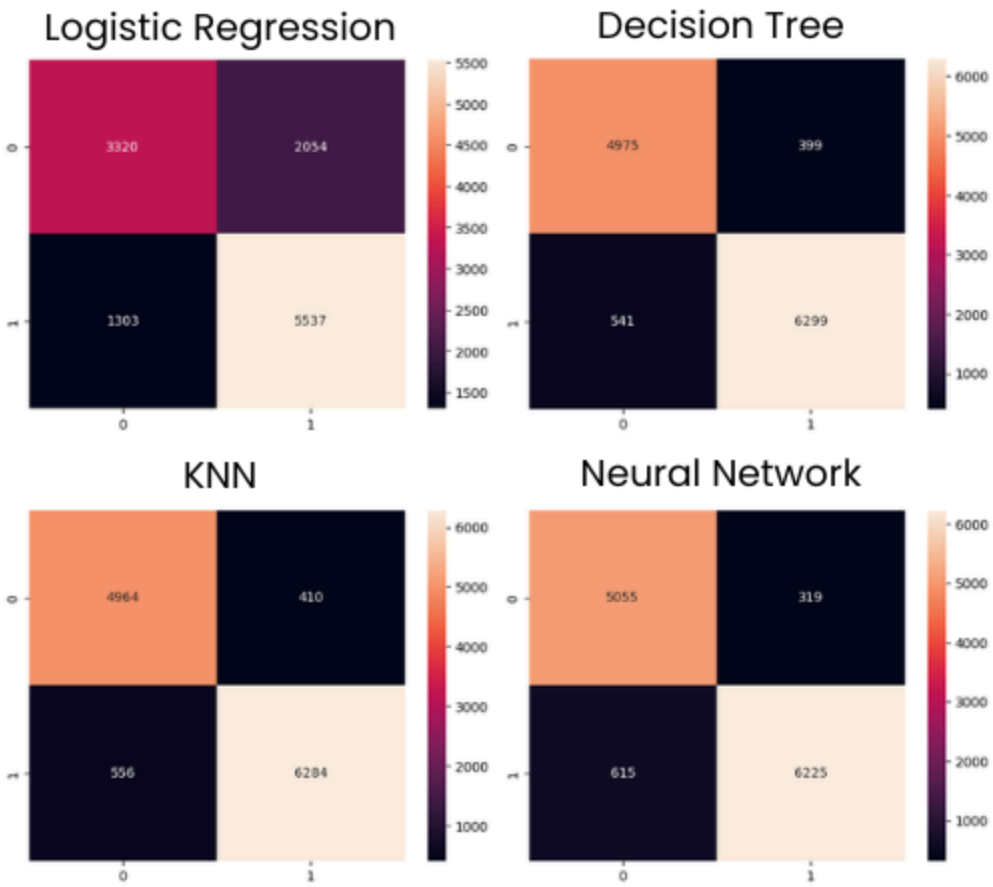
The following table summarizes the performance metrics based on the test data (12,214 samples). High recall for the poisonous class is critical to avoid dangerous misclassifications.

Model	Accuracy	Weighted Avg Precision	Weighted Avg Recall	Weighted Avg F1-Score
Logistic Regression	72.52%	0.72	0.73	0.72
KNN	92.09%	0.92	0.92	0.92
Neural Network	92.35%	0.92	0.92	0.92
Decision Tree	92.30%	0.92	0.92	0.92

Confusion Matrix Analysis

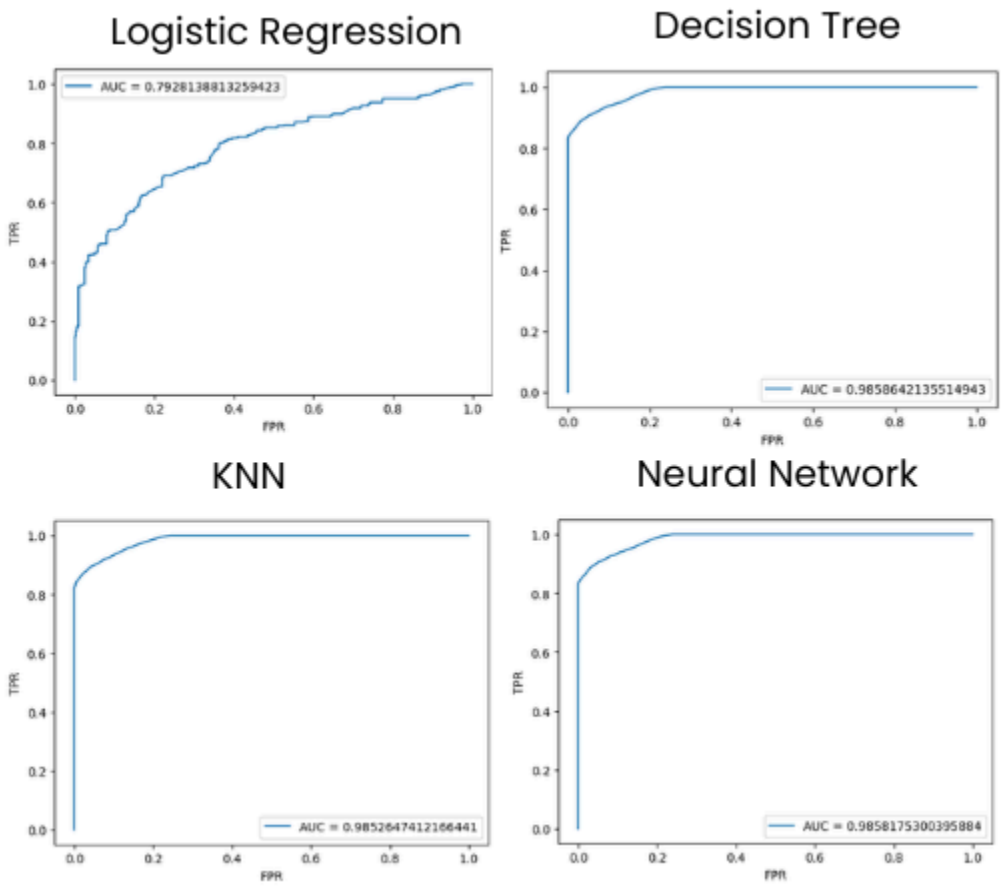
The confusion matrices provide a granular view of model errors:

- Tree-based Models: The matrices show nearly all samples correctly classified on the main diagonal, with negligible misclassifications.
- Logistic Regression: Displayed a higher number of False Positives and False Negatives, confirming it struggles more with the dataset's complexity compared to ensemble methods.



AUC Score and ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) score provide a comprehensive measure of each model's ability to distinguish between edible (Class 0) and poisonous (Class 1) mushrooms.



- Decision Tree (DT): Achieved the highest performance with an AUC of 0.985864. The curve closely hugs the top-left corner, indicating a nearly perfect ability to separate the two classes with minimal error.

- Neural Network (NN): Followed extremely closely with an AUC of 0.985818. The visual overlap of the ROC curve with the Decision Tree highlights its robust predictive power for this dataset.
- K-Nearest Neighbors (KNN): Demonstrated strong classification capability with an AUC of 0.9853. While slightly lower than the tree and network models, it remains an highly effective classifier.
- Logistic Regression: Showed a significantly lower AUC of 0.79. The curve is noticeably flatter and further from the top-left corner, confirming that this linear model struggles more than the others to accurately categorize mushrooms across different probability thresholds.

The high AUC scores (above 0.98) for the Decision Tree, Neural Network, and KNN confirm that these models are exceptionally well-suited for this classification task, whereas Logistic Regression is considerably less reliable for identifying mushroom toxicity in this specific dataset.

Conclusion

The analysis reveals that mushroom toxicity depends on complex, non-linear relationships between physical traits. While the linear Logistic Regression model struggled with an accuracy of 73% and an AUC of 0.79, non-linear models—Decision Tree, KNN, and Neural Network—all achieved a superior accuracy of 92% and AUC scores above 0.98.

Project Findings:

- Model Performance: The Decision Tree is the most practical choice, offering high predictive power (AUC: 0.9858) alongside clear interpretability for biological classification.
- Data Insights: The 8% error rate in top models suggests "noise" in the dataset where edible and poisonous mushrooms share nearly identical features, requiring deep pattern recognition rather than simple linear separation.

Challenges:

- Preprocessing: Managing extensive categorical data and significant missing values (e.g., veil-type) was critical to maintaining model integrity.
- Optimization: Fine-tuning the Neural Network and balancing computational costs against accuracy was a key technical hurdle.

In summary, tree-based and network models provide the most reliable framework for food safety applications, significantly reducing the risk of dangerous misclassifications compared to traditional statistical methods.